Week 4: Planning a project

AI Development Course

Tested with Claude Code 2.1.296, mattpocock-skills 1.2.3, superpowers 6.4.1 and AI-DLC 2.11.0 on Ubuntu Linux (in a container), 12 October 2026, without signing in. Everything inside a signed-in session, on GitHub, or on a Mac or in Windows (WSL) is being tested on real Mac and Windows laptops.

Today

- Why plan first
- Pick your idea: the too-big test, and keeping sensitive data out
- Your project repo, private
- Plan mode, then `/grill-me`
- `PLAN.md`, through a pull request
- Two other ways to plan (demo): Superpowers and AWS AI-DLC
- The assignment, and next week

Why plan first

Claude Code builds exactly what you said, which is rarely what you meant.

Layer	What it does	You get
Plan mode	Reads and thinks; can't change files	A first draft, on screen
`/grill-me`	Questions your plan, round by round	Sharper decisions
`PLAN.md`	The decisions, in a fixed outline, merged by PR	The plan, saved in your repo

Pick your idea

A good idea:

- does **one job** for **one kind of person** (you count)
- fits in one sentence: "It takes ___ and gives me ___"
- uses data you can **type in, make up, or are allowed to use**
- has **3 to 5 must-haves**
- can be shown working in **3 minutes** in week 14

Bonus: a part where AI could help later (summarise, sort, draft).

The too-big test

Too big if it's "a platform", needs several kinds of users or logins, takes payments, copies live data from websites, or uses other people's private data.

Too big	Just right
An app that runs my whole small business	Weekly sales spreadsheet in, one-page summary out
A tutor and student marketplace	A study planner for my exam dates
An AI financial adviser	A budget checker with my own limits (no bank connection)
A real estate investing platform	A rental deal screener for one listing (class example)

Example ideas, about the right size

1. Rental deal screener (the class example, in `example-project/`)
2. Freelance quote calculator
3. Club event budget planner
4. Sales summary from a spreadsheet export
5. Job application tracker with follow-up drafts
6. Product description writer for a small shop
7. Café reorder checker
8. Customer feedback sorter (public or made-up reviews)

Keep sensitive personal data out

Never in your project:

- health or medical information
- account, card or ID numbers
- other people's private information: customers, employees, classmates' grades
- passwords, keys, sign-in codes
- anything you aren't allowed to use

Use **made-up**, **public**, or your own non-sensitive data. (Again in week 13.)

Your project repo (being tested)

In your terminal:

```
cd ~ && mkdir semester-project && cd semester-project && claude
```

`Shift + Tab` until `manual mode on`. Then ask Claude Code to create `README.md` with your idea, commit it, and:

```
gh repo create semester-project --private --source=. --remote=origin --push
```

Before you say Yes, look for `--private`. Plain **Yes** for git, always.

Check github.com: **your** username, and the **Private** label.

Invite your instructor (being tested)

On github.com, as in week 3:

1. Your repo, then **Settings**
2. **Collaborators**, then **Add people**
3. Type `vasuchettyphd`, select it
4. **Add vasuchettyphd to semester-project**

Don't make the repo public. The invitation is how your instructor gets in.

Plan mode (being tested)

`Shift + Tab` until the bottom line says `plan mode on`:

```
auto mode on -> manual mode on -> accept edits on -> plan mode on
```

Or type `/plan` before one prompt.

```
I want to build this for my semester project: "...". I'm not a
programmer, and I have about eight weeks. Don't build anything yet.
Propose a plan: who it's for, the 3 to 5 must-haves, what to leave
out, what data it uses, and the first three build steps.
```

Don't approve it for building

Claude shows the plan and asks how to proceed. Labels vary:

Choice	Means
"Yes, and use auto mode" / "Yes, auto-accept edits"	**start building**
"Yes, manually approve edits"	**start building**
"No, keep planning"	stay in plan mode, tell Claude what to change

This week: **No, keep planning**. Happy with it? Then `Shift + Tab` until `manual mode on`.

Claude Code keeps its own copy in ~/.claude/plans/. Your PLAN.md goes in your repo (step 6).

Install `/grill-me` (mattpocock-skills) (being tested)

In Claude Code:

```
/plugin install mattpocock-skills@claude-plugins-official
```

Choose **Install for you (user scope)**. Asked to? `/reload-plugins`.

Or in your terminal (from our test):

```
$ claude plugin install mattpocock-skills@claude-plugins-official
✔ Successfully installed plugin: mattpocock-skills@claude-plugins-official (scope: user)
```

"Not found in marketplace"? Add the official marketplace first (README, step 5a).

Check: type `/grill` and look for `/grill-me`.

Get grilled (being tested)

```
/grill-me Grill me on the plan you just proposed for my semester project.
```

Not in the list? `/mattpocock-skills:grill-me`

- **Rounds** of **numbered questions**, each with a **recommended answer**
- Reply by number: `1 yes. 2 no, only me. 3 I don't know yet.`
- Facts are Claude's job; **decisions are yours**. Push back.
- It writes **no files**. An edit prompt? Say **No**.

Note your **three hardest questions** and your answers.

How it ends

It ends when nothing is left open, and **you** confirm you agree:

```
Yes, we agree.
```

Out of time, or going in circles?

```
Stop grilling, summarise the decisions so far
```

Anything undecided goes under **Open questions**.

Stopping before it ends (end of class)? Copy the summary into your note. Next time: `claude` in `~/semester-project`, `/grill-me`, paste it, add `Carry on grilling me from here.`

`PLAN.md`: the fixed outline

```
# Plan: <your project's name>
## Goal
## Who it's for
## Must-haves
## Not in this version
## Data it uses
## First three build steps
## Open questions
```

Be specific. "Easy to use" is vague. "Shows the week's total sales in dollars, and the top 3 products" is specific.

Worked example: `example-project/PLAN.md` (the deal screener).

Branch, pull request, merge (being tested)

```
Make a new branch called add-plan. On it, write PLAN.md from our
grilling session, with exactly these seven headings, in order: ...
Use my answers. Be specific. Show it to me before you commit.
```

Then: commit, push, `gh pr create` ("Add project plan").

On GitHub: **Files changed**, then **Merge pull request**, then **Confirm merge**.

Back in Claude Code: "Switch back to main and pull the latest changes from GitHub."

Two other ways to plan (demo, optional)

	Plan mode + /grill-me	Superpowers	AWS AI-DLC
How it plans	Draft, then rounds of numbered questions	One question at a time, 2-3 approaches, approve section by section	Phases, one stage at a time, approval gates
Produces	Nothing; you ask for `PLAN.md`	Design and plan files in `docs/superpowers/`	Records in `aidlc/`, ~40 skills in `.claude/`
Use it for	Deciding what to build	Building in small, tested steps	Team software projects

Next week you build with Superpowers.

Trying them yourself (optional)

Only **after your `PLAN.md` PR is merged**, in a **throwaway folder**:

```
mkdir -p ~/planning-sandbox && cd ~/planning-sandbox && git init && claude
```

Superpowers: `/plugin install superpowers@claude-plugins-official`, then `/superpowers:brainstorming ...`. User scope means it's on in **every** folder, so remove it afterwards, in your terminal: `claude plugin uninstall superpowers@claude-plugins-official` (or `/plugin uninstall superpowers@claude-plugins-official` in Claude Code, being tested). If Claude Code is open, type `/reload-plugins`.

AI-DLC (Windows: the Ubuntu window). It gives Claude a very broad edit permission (exact reach being tested). Use its own folder:

```
mkdir -p ~/aidlc-sandbox && cd ~/aidlc-sandbox && git init
curl -fsSL https://github.com/awslabs/aidlc-workflows/releases/latest/download/install.sh | sh
aidlc config --harness claude && aidlc doctor
```

Then /aidlc ... in Claude Code. Runs are being tested.

Assignment

One document with:

- Your **private** project repo's address, a screenshot showing **Private**, `vasuchettyphd` added
- The address of the **merged** PR that adds `PLAN.md`
- Your **three hardest** `/grill-me` **questions**, your answers, and how one changed your plan

Criterion	Points
Repo private (screenshot) and collaborator added	3
`PLAN.md`: outline (1), specific and no sensitive data (1), merged by PR (2)	4
Three grilling questions, answers, and what changed	3

Next week: Building with AI

You build the **first steps of your `PLAN.md`**, with Superpowers.

Bring:

- your merged `PLAN.md`
- your open questions, answered if you can